

Performance in planning — smart systems for the access network

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Improvements in system technologies continue apace — with new developments in software methods arguably outstripping even the breakneck progress in hardware power described by ‘Moore’s Law’. The software developments are timely, since they offer solutions to the urgent problems of increasing customer demands, complexity of network technologies, and more competitive markets. The increasing availability of cheap computing power, with developments in object orientation and artificial intelligence, offer the possibility of a more holistic approach to network performance planning and management. The most fundamental aspects of network performance (functionality, reliability, and cost) can now feasibly be considered simultaneously and interactively. Common, comprehensive network models are within sight, which could be utilised by network planners, installers, repair staff, marketers and at point-of-sale. The resulting improvements in productivity and service quality could be immense. Specific examples of ‘intelligent’ access network planning tools, produced using rapid application development techniques within BT, are described to illustrate their potential.

1. Introduction

The pressures on telecommunications service providers are now well recognised, and changes in the technologies and markets are increasingly dynamic. Market liberalisation is well-advanced in the USA and Europe, and is being implemented almost everywhere else. Customer expectations of service quality, availability, and value are ever-increasing [1]. Growth in Internet and data transfer demand has increased stress on core network capacities, challenged access network capabilities, and created new niches for market entrants. Continued technological progress, particularly in semiconductor integrated chip development, is creating exciting opportunities to provide new broadband and multimedia services. This technological trend, in fact, is the root cause of most of the opportunities (and problems) confronting the converging telecommunications and information technology industries.

It is easy to forget, but the prediction made in 1968 by Gordon Moore continues to apply — an approximate doubling of semiconductor chip density every 12–18 months. From ‘Moore’s Law’ flows the annual increase in personal computer (PC) processing speed and memory we have grown to expect, and also such developments as adaptive terminal equipment, crucial to the asymmetric digital subscriber line (ADSL) systems which can deliver video bandwidths to residential customers over ordinary telephone wires. Optical fibre, co-axial cable and radio technologies compete for the main market segments, and new possibilities such as Internet telephony are challenging traditional market models. If the opportunities created by

these changes are huge, then so are the corresponding problems of planning and managing the increasingly complex networks required. Fortunately advances in software engineering methodologies have at least matched the spectacular developments in hardware — object orientation, rapid application development (RAD) techniques, distributed database management methods, and expert systems offer the possibility to plan and manage future complex networks, and markets, more efficiently than the relatively simple networks of the past.

2. Network performance and planning — processes and opportunities

Currently, a number of fundamental changes are affecting how the access network is planned and new services subsequently delivered at the strategic, tactical and detailed levels. Firstly, both the services offered and the demand profile for these services have seen a dramatic change in terms of differentiation and segmentation. Secondly, the deregulation of the telecommunications market within the UK has introduced fierce competition from other network operators, with over 150 licences being issued to new operators since 1991. Thirdly, BT has continued to extend its business activity into global markets primarily through collaborative joint ventures, often facing near-to-medium term deployment scenarios quite different from those in the UK. Finally, new technology continues to extend the performance envelope of the access network, making it possible to plan for and to deliver a wider range of

‘datawave’ and broadband services. At the front end of each of these diverse activities is the planning process. It is at this point that the future risk to any network infrastructure build proposals can be minimised. Quite simply, getting it wrong at this stage has a massive impact on subsequent return on investment and hence continued profitability. Therefore, at the planning phase of the access network, a number of key questions need to be addressed in order to meet time and cost targets:

- where investment should be made,
- the impact of competition,
- emergence of new service offerings together with any constraints on the underlying service provisioning platform,
- what technology and quantities of plant should be deployed,
- in what time periods the deployment should be made.

These are complex issues, and often a number of techno-economic scenarios must be considered. Additionally, these decisions must often be made on a continuous basis to tight deadlines. These factors naturally lead to the need for tools that will facilitate the planning process through at least partial automation. This paper will describe work which has addressed the issues of access network planning at the strategic, tactical and detailed levels. While dealing broadly with many aspects of the planning process, the paper will focus on a specific example of how innovative methods from the field of artificial intelligence (AI) can be applied to augment the skills of a network planner in the area of greenfield access network planning.

The process of designing and planning new network implementations can be considered the first stage of the network management life cycle. Network planning is a skilled, labour-intensive process in which many different aspects of the network performance must be considered. For example, the basic functional performance of the network must be within design limits (e.g. transmission loss tolerances) and the link bandwidths must be dimensioned to handle forecast traffic and signalling overheads. The overall network availability must be calculated, based on estimated component reliabilities, and redundancy must be introduced if required. The network must also be designed for minimum cost, which may mean a relatively simple calculation of installed cost if a single technology is considered, but could mean comparisons of whole-life cost if technology comparisons are required. Traditionally, these various aspects are considered to a great extent separately and sometimes even sequentially, in an attempt to make manageable the complexities involved. Thus there has evolved many different process steps, planning rules, and species of network planners, investment appraisers,

strategic planners, tactical and detail planners, copper, radio and fibre network specialists, traffic modellers, and reliability engineers. With all these planning steps and considerations, even experienced human experts have little chance of efficiently designing truly optimal networks. It is clear that, against the background of increasing demands on the performance of the networks and of the planners, network planning is a discipline full of opportunities.

3. Optimising ‘total’ performance

The way forward is to exploit those very technologies that are giving rise to the problems of increased network complexity and customer demands. Computing power increasingly offers the opportunity to break away from the arbitrary fragmentations of traditional network planning, towards a more comprehensive and holistic approach. A reasonable definition of the domain of ‘total’ network performance that must be navigated is the functionality/reliability/minimum cost triangle. On this ‘map’ [2], whole-life cost is simply the overlap of cost and reliability, and optimum bandwidth (capacity) dimensioning can be considered to be a trade-off between functionality and cost. The three fundamental aspects of network performance — function, reliability, cost — can now feasibly be considered simultaneously and interactively, vastly improving the possibility of obtaining optimal design solutions, even for complex networks. Unified network models, capturing the ‘total’ performance characteristics of the network, could become a powerful tool, not just for network planners and managers, but also operational and maintenance engineers, sales people, and marketers.

If this vision of more unified and universally useful network models is to be realised, then of course more than just the performance of the network ‘machine’ must be considered. Extra types of data, such as customer socio-economics, and billing or revenue data, could be required to segment and address the target markets. Also, the effect of human factors on network performance can be important, for example, most telecommunications companies are aware of the deleterious effect of human intrusion in the access network. Increasingly even these complexities can be, and must be, considered in planning the competitive networks of the future [3].

4. Systems developments and trends

How then are we to best exploit the increasing power of our computing hardware and software, to move towards comprehensive network planning and ‘total’ performance? Several trends have clearly become important in the last few years. Object-oriented programming and data structures with their inherent characteristics of encapsulation, inheritance and polymorphism are ideally suited to the efficient modelling of telecommunications networks [4]. ‘Layered’ network data models which describe the hierarchy and containment of network objects (e.g. duct-

cable-bearer-channel) are crucial to any powerful planning model. A commonly shared, readily accessible cache of network plant and performance data is essential if wasteful data duplication is to be avoided. User-friendly interfaces are as always required, and this usually means at least a graphical user interface (GUI) and often needs a geographical information system (GIS) approach. A general systems architecture which shares data, but allows a diversity of client applications and presentation layers, offers the best compromise between data integrity and application quality [5]. Rapid application development techniques are increasingly powerful, and the resulting applications, if modular and coupled with 'core' data, can produce powerful results. The requirements can be summarised as a set of network performance models and planning tools which are:

- modular,
- distributed,
- interconnected,
- based on shared data,
- easy to access and use,
- holistic and comprehensive.

'Thick' client systems, sharing data via intranets, are increasingly likely to meet these requirements. Expert system applications are being developed which can solve difficult network optimisation problems in a few seconds, running on a desktop PC [6].

5. Examples of developmental tools and systems

Several examples of such recent developments, in use within BT, illustrate the possibilities. In varying degrees, they are expert systems — they embody the skills and rules employed by skilled human planners, and their impact on planning productivity and quality is proving to be extremely significant.

5.1 BECS

The first, BECS (Budgetary Estimate Calculation System), is a medium-size (30 000 line) Visual Basic (VB) application which allows planners to estimate quickly the cost of 'diversionary works', i.e. the cost of moving and re-arranging existing external network plant and cables (for example, to accommodate new building developments or roads). It is menu-driven, very easy to use, and based on shared models and data which detail the external plant costs and installation times. A key feature of BECS is that while the application software is widely distributed and runs on users' desktop PCs, the 'core' data on costs and installation times is updated, controlled and distributed from a central point, ensuring enterprise-wide consistency.

5.2 IDACS

IDACS is an investment decision and control system for the access network. It aims to implement the principles of customer segmentation and nominates priorities for investment to cater for growth in customer demand or exhausted network capacity. Specifically, it is being developed to abstract data from a large number of enterprise-wide databases, present this in coherent form to the planner, and suggest capacity growth priorities based on various demand criteria. Again, it is a VB application with network links to BT mainframe and client/server systems containing the core data.

5.3 TASC

TASC is a graphical, menu-driven, semi-automatic planning tool for the production of authorised 'standard' cases, i.e. the approval of financial business cases for access network planning schemes. It is initially aimed at simplifying the planning and cost-benefit analysis of the provision of passive optical networks (PONs) and conventional copper networks. The cost and net present value of the PON and copper schemes is calculated and apportioned over time and customers, based on input circuit demand forecasts and outline costs, and bills of material are produced for both schemes. The most labour-intensive steps in the production of the standard financial approval case (FAC) calculations and document production are completely automated, and the result is major time-savings in FAC submission preparation (e.g. 2 days work can be reduced to 15 minutes). Just as importantly, the improved productivity and standardisation of processes leads to more cost-effective targeting of PON technology on the best-suited customer areas. Development of TASC was completed in 18 man-months, and the tool has been produced in Delphi to obtain the code-reuse benefits of fully object-oriented Visual Pascal.

5.4 GenOSys

Lastly, the GenOSys genetic optimisation system is a sophisticated tool, again coded in Delphi, which employs genetic algorithms to minimise the cost of greenfield copper access network developments against a well-defined set of possible geographical locations and network planning rules. Genetic algorithms, employing the principles of selecting the fittest from many solutions 'bred' over numerous generations, are computationally efficient and very resistant to entrapment in the 'false optima' which abound in complex solution spaces. Again the system has proved its worth in massive improvements in planning productivity, mainly as a result of the easy-to-use GIS. The genetic algorithm optimisation engine effectively guarantees that the lowest-cost network solution is planned, and is capable of optimising highly complex 'objective functions' which could describe any aspects of network 'total' performance.

Approximately 15 man-years of R&D has been invested in this system, but the final applications development was completed in around 4 man-years. The next section describes in more detail how GenOSys has embodied innovative methods in software engineering and artificial intelligence — object-oriented technology, genetic algorithms and rule-enforcement inference engines — and applied them to the practical problem of designing copper greenfield networks.

6. An expert system for greenfield planning — GenOSys

This section describes how innovative methods have been applied to the problem of automating the copper greenfield planning process within BT in the UK. The work has involved collaboration between BT Laboratories and Sunderland University and is based on leading-edge techniques in artificial intelligence, specifically the development of customised genetic algorithms well-suited to solving access planning problems quickly on desktop PCs. AI techniques offer a number of advantages; in particular they can address ill-defined problems (e.g. missing or incoherent data and mathematically intractable problems), solve problems to which conventional algorithms cannot be realistically applied (e.g. very large search spaces), and in some cases learn from experience. The set of tools described here has been successfully proven in test-bed trials and the business benefits projected to accrue from their field deployment is about £10 million per annum, or about 10% of the total annual investment made in copper greenfield networks.

6.1 The detailed copper greenfield planning process

In BT the primary method of providing service from a local exchange to the customer is by means of copper pairs. The continuing investment in copper technology — particularly for new housing development — makes it desirable to automate as many parts of the infrastructure design process as possible. Currently, the entire area of copper greenfield activity represents an annual investment of about £100 million per annum to the company. A simplified diagram of the access network is shown in Fig 1.

Most of the design activity carried out to create a scheme for a greenfield network is done between the primary crossconnection point (PCP) (made available at the entry point of the scheme) and each customer (the D-side network). This activity takes place across and within four layers. The site developer's initial geography and actions define the civil works layer, e.g. the position of customers and the layout of site trenching. The duct layer is contained within the civil works layer and consists of the duct pipes and intermediate boxes that serve to connect sections of ducting together. Enclosed within the duct layer is the cable layer. In this layer, groups of bearer pairs are aggregated

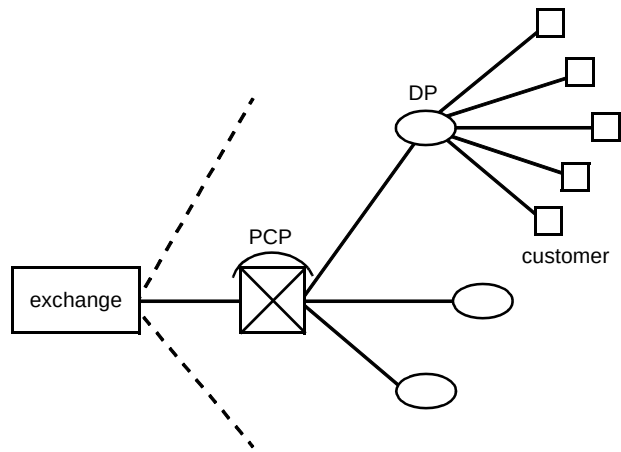


Fig 1 A simplified view of the access network.

into cable sheaths. Within the cable sheaths are the bearers themselves which deliver services to the connected customers. Figure 2 illustrates an overview of the copper greenfield planning process and the relationship to each of the layers highlighted above. It begins with the site plans of the proposed site being received from a developer (typically on an A0 sheet of paper). Upon this plan, the network planner would lay out a duct network, and submit it for consideration to the developer. If the proposal is acceptable, a cable network can then be produced. Additionally, a jointing schedule is prepared to specify how each working pair on every cable is to be jointed back from each customer to the exchange. The duct network is comprised of a spine network and a lead-in network. The spine network consists of 90 mm diameter duct laid between the entry point to the site — usually a PCP — and access points which will subsequently contain the flexibility or distribution points (DPs) of the scheme. The lead-in network consists of 49 mm diameter duct and is used to establish connections between the DP housings and the site customers. Between the PCP and each DP, the main spine duct comprises sections of ducting which run between access points on the network. These access points are usually boxes which can either be pre-fabricated or built on site by the developer and they enable junctions to be created on the duct network. Junctions serve two major functions in the duct network. Firstly, they allow cables to be introduced into any part of the duct network. Secondly, they permit the introduction of points of flexibility or jointing into cables that pass through or are terminated at a box. Between the DP housings and each customer, the lead-in network consists of 49 mm diameter duct sections and optional branch elements (duct tees). Tees obviate the need to run individual sections of duct between each customer and the serving DP. Instead, cable can be introduced by rodding from each customer through an intervening duct tee up to the box which houses the serving DP. Figure 3 illustrates this process. There are, however, constraints on the use of duct tees which will be discussed further in section 6.2. In both the spine and lead-in networks, changes of duct link direction are achieved by using a variety of duct bends.

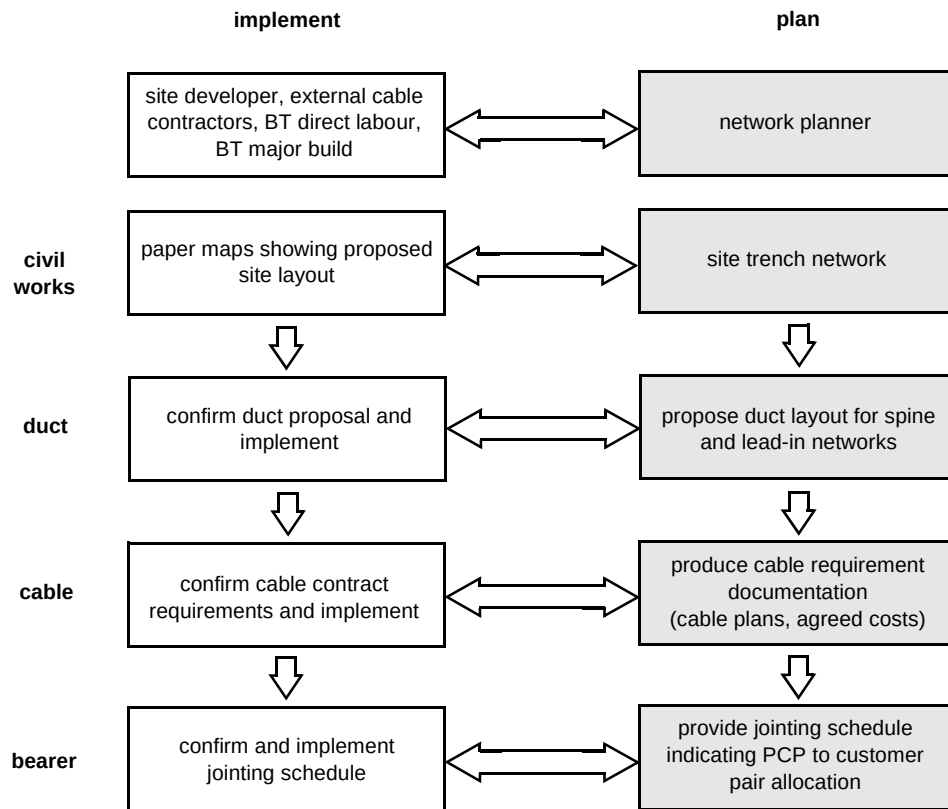


Fig 2 A layered view of the copper greenfield planning process.

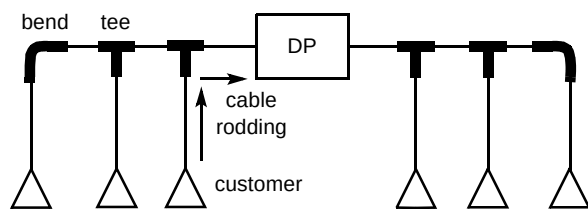


Fig 3 Duct tees in the lead-in network.

Once a duct network has been agreed, a cable network can be introduced. The cable network also consists of two parts — the spine network which is located between the PCP and each DP, and the lead-in network which is located between each DP and customer. In reality, because cables are manufactured in discrete sizes, with 100 pairs being the largest deployed size in the D-side network, a number of cable networks may exist, with each network serving up to a maximum of 100 pairs. Beginning from the customer edges of the duct network, pairs are allocated based on the predicted level of demand per customer. At this point, a decision about the number of customers that will be served (and therefore the total number of copper pairs) and the position of the serving DP needs to be made. DPs are usually sited at the most central box in the cluster of customers. This position tends to minimise the run of cables between the DP and each customer. Once all the DP pair counts and positions are determined, the design of the spine network cable can begin (Fig 4). The goal in the design of the spine network is to provide the smallest number of

cables to provide service to groups of DPs. In order to do this, the DPs are grouped in such a way that the aggregate number of pairs for each set of DPs does not exceed 100. Each cable that provides service to a DP group has its outer sheath exposed at the box which contains a DP, and then

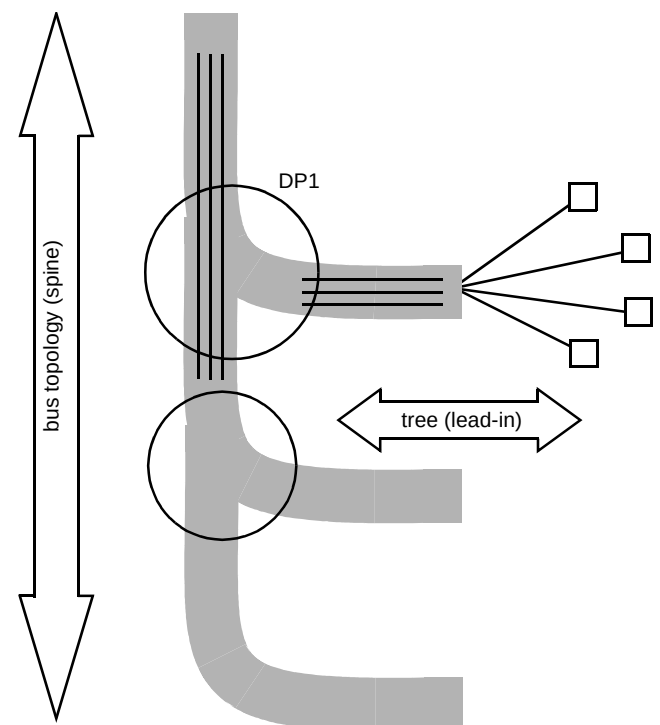


Fig 4 Spine and lead-in topologies.

only the required number of pairs are exposed for jointing at the DP. Thus the cable spine network forms a logical bus. At the bearer level, of course, the configuration is a tree.

Upon completion of the cable network, a jointing schedule needs to be prepared. This document defines how connectivity at the bearer level is achieved for a given network. In order to minimise the risk of damage, the pairs on the outermost part of a cable are allocated to the most upstream DPs or customers, with the assignment of the innermost pairs being to the more downstream DPs or customers. Figure 5 is a simplified illustration of this process.

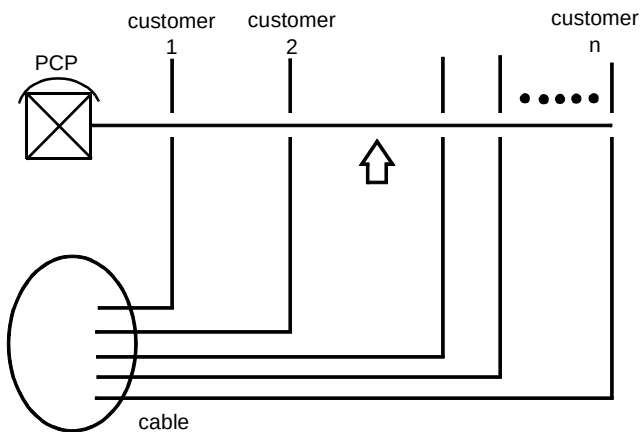


Fig 5 Allocation of cable bearers on basis of distance from PCP.

6.2 Objectives of copper greenfield automated planning

The overall objective is to create a system which will autonomously design copper greenfield network infrastructures that satisfy engineering and cost constraints. Once the civil works layer of the infrastructure has been specified by a network planner, the autoplanning system should be able to synthesise the required network infrastructure at the duct, cable and bearer layers. The autoplanning system should further be capable of producing any documentation and estimates of infrastructure cost in each design layer required by the network planner, the site developer and any external contractors in an acceptable form. Essentially, the documentation produced should meet the standards required for internal record keeping and also be able to be used as the basis for external contracts with site developers and external contractors. The auto-generated design should be cost-effective to implement and meet any engineering constraints imposed by the type of network plant being deployed. Furthermore, any infrastructure designs created by the system should be produced with a minimum of intervention from the network planner. Finally, the application of the system to greenfield infrastructure design should deliver significant benefits in terms of the actual amount of time spent on doing the design work.

6.3 System requirements for a copper greenfield auto-planning tool

Given the existing view of the generic copper greenfield planning process, and the objectives which need to be met by an automated design process, this section will examine in greater detail the requirements, desirable attributes, key components and business benefits of such a system.

- The system should save design time

Planners work to a set of planning rules which ensure that designs are technologically feasible and can be produced within a reasonable time. Given the pressure of job planning volumes and the time taken to explore alternatives, very few alternative infrastructure scenarios are usually considered in the manual planning process. A system which can explore a large number of potential infrastructure options within a relatively short time, and assure consistency across explored options, can yield considerable savings in time. This is particularly true as design progresses from the specification of infrastructure to the detailed costing and documentation stages. Therefore, even when the planner must override significant portions of an auto-generated design, the time saved in recomputing costs and reworking documentation is essentially only a few mouse clicks away all the time.

- A feasible and cost-effective infrastructure design is required

This is a central requirement for an automated planning system. It is a challenging task because of the number and kinds of parameters as well as the mix of engineering and/or policy constraints which must be factored into the creation of candidate infrastructures by the system. For instance, generating a cost-effective cable layer may lead to a duct layer network failing to meet engineering constraints. Changing the civil works layer specification (done by the planner) may stop the system from meeting duct layer policy constraints. Many feasible designs may not be cost effective. However, this type of task is well suited to the methods of artificial intelligence and section 6.5 will describe how conflicting restraints and large parameter mixes have been handled by the system.

- Optional and progressive automation is desirable

Responsibility for the outputs of an automatically generated scheme must be assumed by a network planner. A highly experienced planner might for instance choose to specify every aspect of a proposal, using the system as a means to save time in the preparation of proposals. Alternatively, such a planner pressed for time might elect to have the system auto-generate the duct and cable layers and then override a number of design decisions made by the system. In this alternative scenario, it may be that the planner is

completely satisfied with the outputs of the system and no further modification of the initially produced design is necessary. Therefore, a system that allows automatically generated designs to be overridden or modified in the light of a planner's experience is desirable. Such a facility allows the proper flexing of automation against the experience of the planner. The challenge is to perform such actions seamlessly and in such a way that, in whatever mode (manual, semi-automatic or fully automatic) the planner chooses to use the tool, exactly the same set of actions are required for interaction.

- System component reuse is desirable

In order to solve similar types of problems in future, it is desirable that the framework upon which the system is constructed lends itself to rapid reuse in related problem areas. For instance, the problem of primary network fibre provisioning is analogous to the problem of copper greenfield spine network provision. Similarly, the problem of service distribution from a central serving facility contains the same essential elements regardless of the medium used to deliver service (for instance radio or fibre). In such cases, the extent to which the similarities in problem domains may be successfully exploited depends on the intrinsic nature of the framework used to express and implement the copper greenfield problem. In the most general case the question is: 'How can the array of subsystems put

together for the purpose of copper greenfield network planning automation be readily deployed in scenarios that require networks to be optimally planned, and design and costing documentation subsequently generated?'

Object technology has been extensively used in the design and subsequent implementation of all the constituent subsystems described in this paper. The steps of meta-class abstraction have been applied at all stages in the continuing life cycle of the system's architectural framework. This has been particularly true of the early stages of the framework. It has subsequently been found that building systems in this way enables the inevitable change in customer requirements to be rapidly and effectively addressed. The core object representation model is presented in the next section.

6.4 System implementation detail

Figure 6 highlights the major subsystems (modules or components) of the automated planning system and their relationships to each other. The design closely mirrors the requirements specified in section 6.2.

A key subsystem of this infrastructure is the communication object-oriented component architecture (COCA). It represents the data model of the access network infrastructure, and is defined by the layered object modelling technology (OMT) representation of Fig 7. In

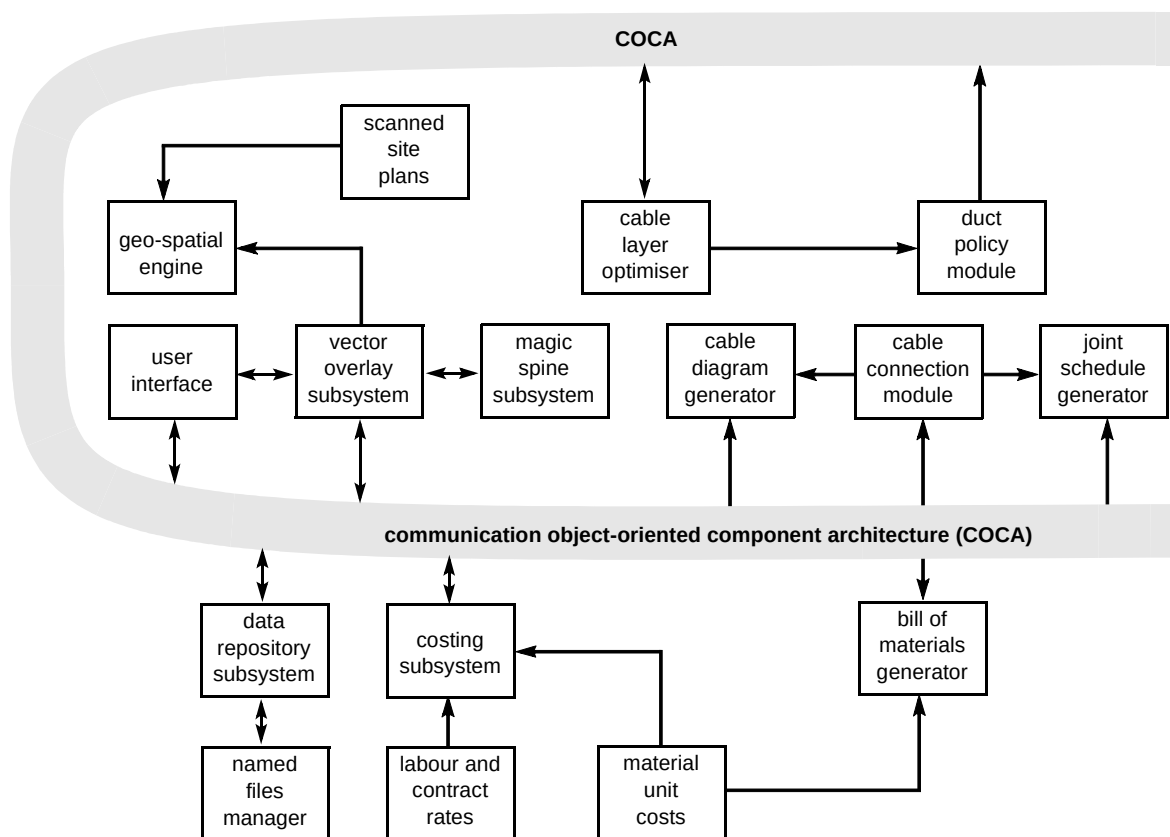


Fig 6 Automated copper greenfield planner — major subsystems.

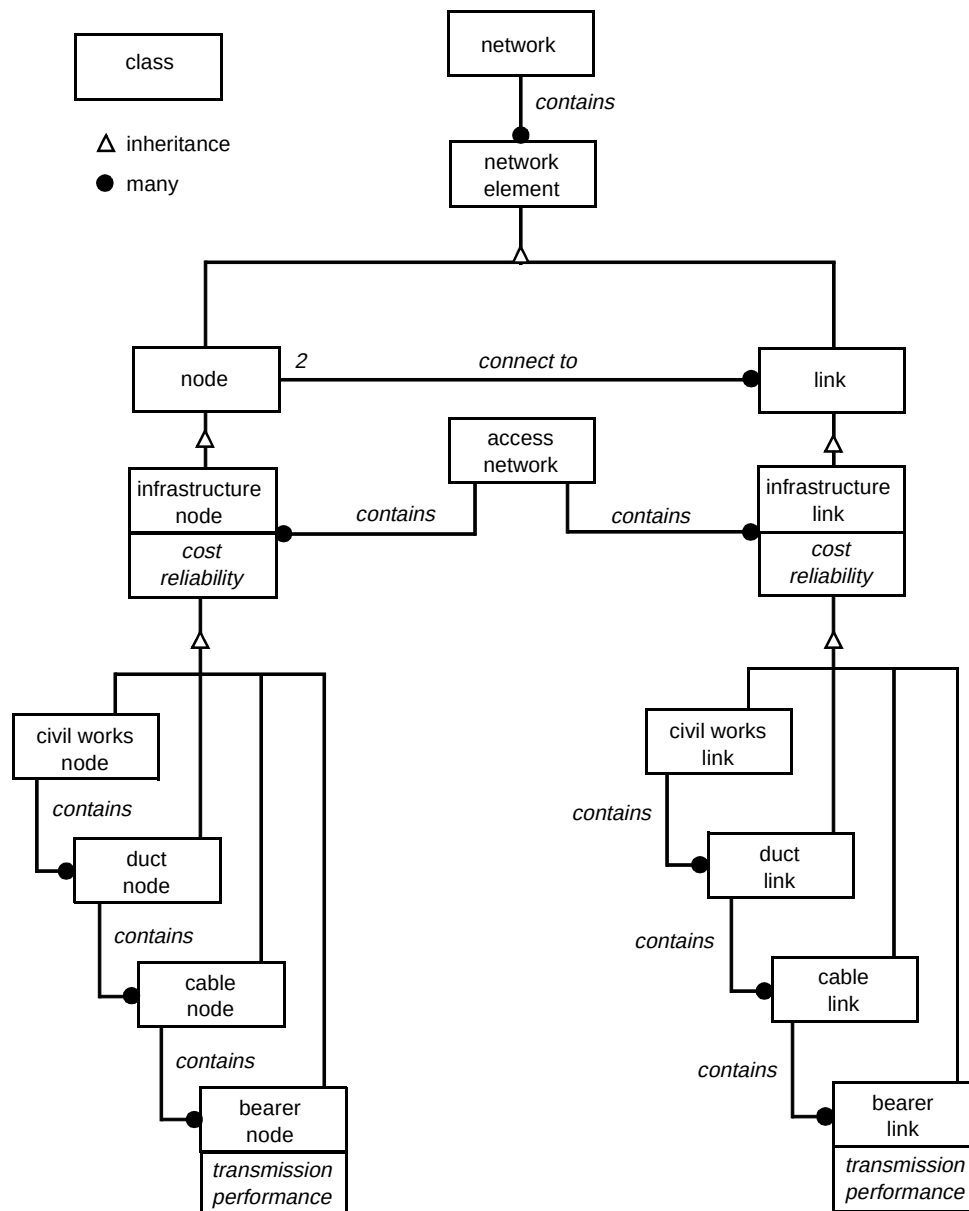


Fig 7 OMT specification of access infrastructure meta-classes.

essence it is a real-world representation of the access network infrastructure in software, reflecting the properties of its fundamental components as well as the layering and connectivity relationships between these components. COCA is central to the GenOSys architecture, providing the means for all the other subsystems to provide functionality and to interact. The geospatial component enables the site plan obtained from the developer to be displayed (after it has been scanned and converted into a raster image), registered against the real world and subsequently manipulated by the planner. The registration process consists of setting a scale, defining a real-world point of reference (for instance, a known spatial position, such as a PCP northing and easting) and setting the north orientation. Against the raster image of the site plan, the vector layer of the scheme can be overlaid using a mix of planner experience and automation. The vector layer is modelled

using a set of meta-classes which serve as a true representation of the plant items used in the access network. Figure 7 illustrates the specification of the network infrastructure abstraction. In concrete terms, the vector layer consists of items on the civil works, duct and cable layers. The civil works layer is entirely placed by the network planner and it consists of the geographical position, demand level (in voice circuits), and type of each customer together with the trench network that provides connectivity between each customer and the site entry point. The trench network consists of track sections along with the intermediate hole or pole positions which enable node access at the duct and cable layers. The duct, cable and bearer layers have been discussed in section 6.1.

Figure 8 highlights the key elements of the automation component. This component is based on the application of

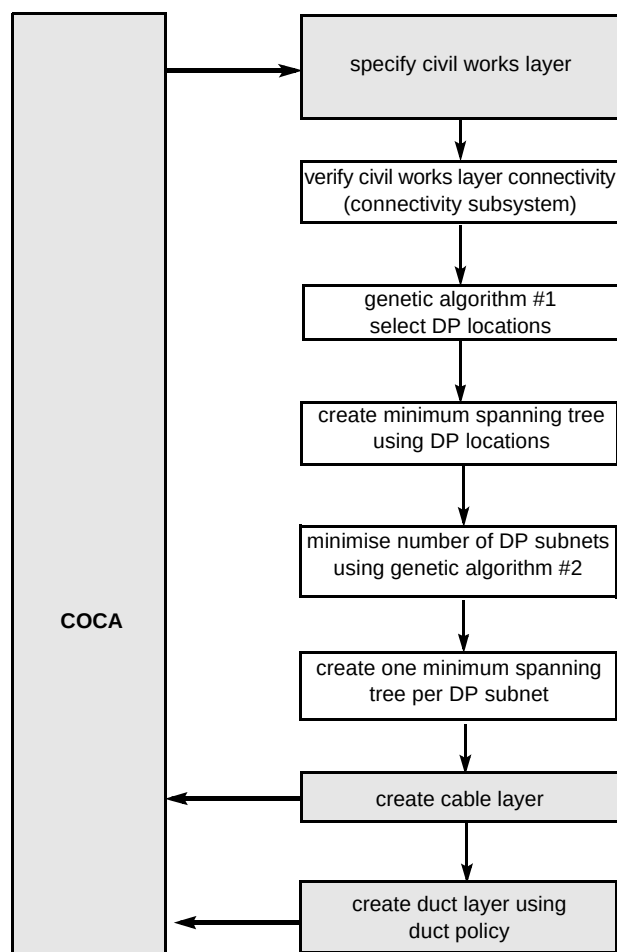


Fig 8 Automation subsystem.

genetic algorithms and graph theory to effect a transformation of the civil works network into a cost-optimised cable layer. The subsequent encapsulation of the cable layer to create a duct layer is done using the policy component. Policy reflects real-world engineering constraints on infrastructure design such as the maximum allowable distances

for pulling cable or the number of in-line duct tees which will permit rodding through duct without damage to cable. This subsystem is explored further in section 6.5.

The costing component generates a summary scheme cost based on the actual cost of each item of network plant used in the infrastructure design. In addition to a bill of materials, the labour costs of installation activity in the duct and cable layer are also taken into account. This component enables either detailed (material and labour for each network element) or summary costs (across the entire scheme) to be examined. Figure 9 illustrates the detail of the costing component.

The documentation component generates hard copy of relevant data from each layer of the infrastructure design into specified formats. It ensures that the standard of the generated documentation is fit for both records and contractual purposes. The key subsystems of the documentation component are the plant on map, cable diagram, and joint schedule generator. These subsystems and the relationships between them have been shown in Fig 6. The data repository component implements a persistent state snapshot of all layer data to and from named files. This component has been designed to be scalable, and will also transfer data between the system and databases located either locally or remotely. Figure 10 illustrates how this could be achieved.

6.5 AI subsystems detail

The automation component is primarily based on the application of genetic algorithms and graph theory to the problem of specifying a duct and cable infrastructure from a civil works layer specified by the network planner. Genetic algorithms [7] are search procedures based on the mechanics of natural selection and natural genetics. They have been used within the system's automation component in two ways — firstly, to select DP locations from the set of civil works nodes in the network, and secondly, to group

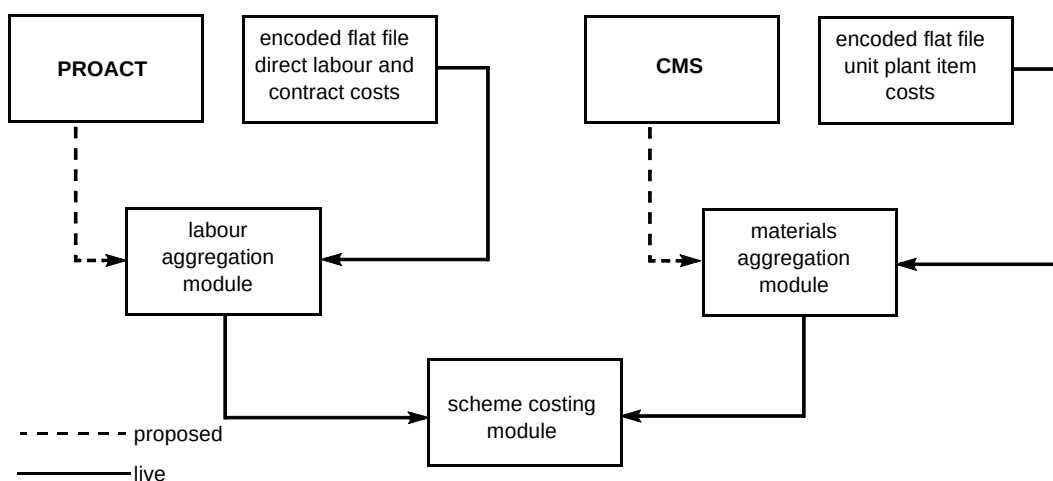


Fig 9 Costing subsystem detail.

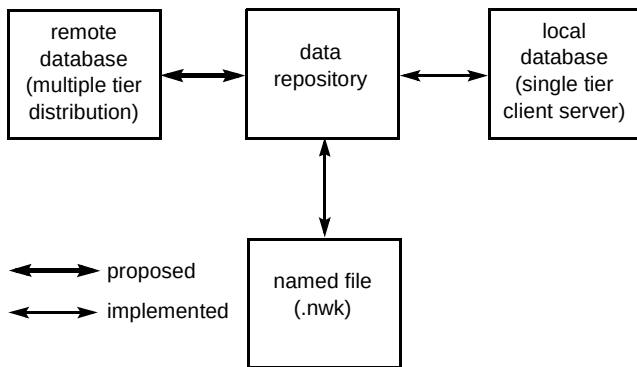


Fig 10 Data module scalability.

these DPs in such a way as to minimise the number of feed cables which are provided from the PCP. Graph theory has primarily been used to determine which customers are most effectively served by DPs on the basis of minimum trench distance. Key to the application of genetic algorithms is the ability to represent the real-world problem or its abstraction as a string of bits (or numbers) known as chromosomes. The search space for the problem consists of the number of distinct patterns that can be formed from the encoded string. For instance for a binary chromosome of length n , the size of the search space is 2^n . Of course, not all of the encoded string patterns are valid in the sense that when decoded they will not form meaningful real-world objects. The task for the genetic algorithm is to efficiently discover which encoded string satisfies a given set of constraints. The algorithm operates on a set of offered solutions (individuals) referred to as a population. Each individual is evaluated by the target (objective) function in order to determine its fitness. Evolution occurs by creating a new population of individuals based on the application of three operators to the current population. The first of these operators is reproduction. An individual is selected from the current population and transferred to the next generation. The probability of an individual being selected for transfer is proportional to its fitness value — a computational analogue of natural selection which ensures the survival of the fittest solutions (see equation (1)).

Probability of reproduction of i th individual =

$$\frac{F_i}{\sum_{k=1}^{pop} F_k} \quad \dots (1)$$

where F_i is the fitness value of the i th individual. The second operator is crossover.

Two individuals, chosen at random, exchange genetic information. As a result, two new individuals are created. The process is illustrated in Fig 11. The last operator is mutation. Mutation creates a random change in a single position of an individual chromosome as shown in Fig 12.

Mutation enables the genetic algorithm to explore new areas of the problem search space, effectively reducing the

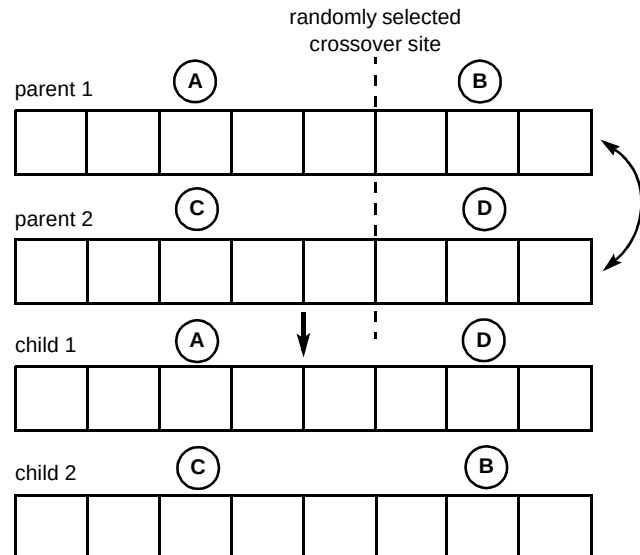


Fig 11 Genetic crossover operator.

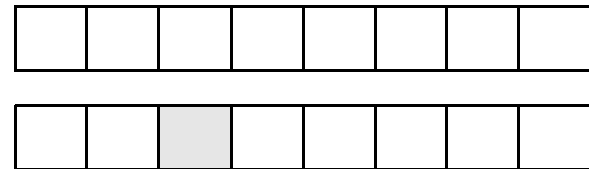


Fig 12 Mutation operator.

tendency to get stuck in a local optimum. From the current population, a new population is evolved using these three operators. Individuals for the new generation are reproduced. Crossover is then applied to randomly selected pairs in order to exchange genetic material. Finally, mutation is randomly introduced. The fitness function then evaluates the fitness of each individual and finally the new generation replaces the old one. From this point the algorithm is repeated. Empirical measurements done within the planning systems team indicate that carrying out this process for a number of generations, equal to about four times the total number of nodes in the civil works layer of the proposed infrastructure, yields high-quality solutions. Furthermore, work by De Jong and Goldberg [7] recommends a high choice of crossover probability, a low mutation probability (roughly in inverse proportion to population size), and a moderate population size (about 50). These guidelines have been followed in the application of genetic algorithms within the copper greenfield planning system. Figure 13 illustrates the evolution process.

6.6 Application of the automated planning system

In order to create a design using the automated system, the network planner follows the modified process illustrated in Fig 14. As far as the site developers, external contractors and direct labour units are concerned, the key change in the process is that the network planner suddenly appears to be significantly more responsive in dealing with proposals. Otherwise the process is unchanged with respect to the type

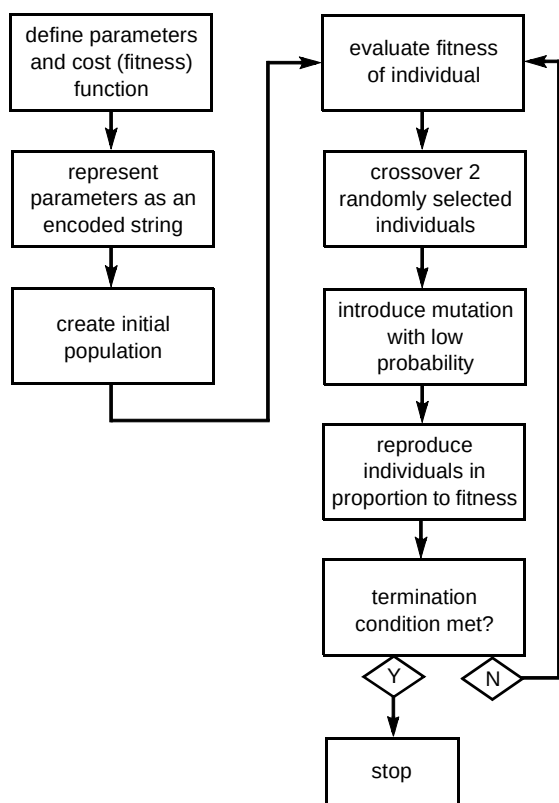


Fig 13 Operation of the genetic algorithm.

and quality of documentation that flows between them and the network planner.

The planner starts with a scanned map of the location. Upon this a civil works (track or trench) network is overlaid using a drag and drop metaphor from a set of icons which represent links and nodes. Current planning rules [8] recommend serving customers on the lead-in network with a higher number of duct tees in order to share the duct costs and to reduce the need for boxes (which are more expensive). Therefore, the planner would place civil works node positions on the frontage area of each spotted customer. Figure 15 illustrates this process.

The ‘magic spine’ component automates this process within the tool. It allows a route to be defined along the pavements and road crossings and then locks spotted customers on to this route. This process is typically between 5 and 8 times faster than placing all the node and intermediate links on the civil works layer. At this point in the process, the planner can invoke automation to complete the scheme or proceed to manually enter the duct and cable layers. If automation is invoked, the AI-based components of the system will synthesise the duct and cable layers of the network as has been described in the preceding sections. Alternatively, the planner can manually enter the cable and duct layers of the network. A prerequisite for automation is that the civil works layer of the network is fully connected. While connectivity algorithms will flag any breaks in this layer, the user is required to visually confirm breaks and close them. To facilitate this, a network overview has been provided within the tool and is illustrated in Fig 16.

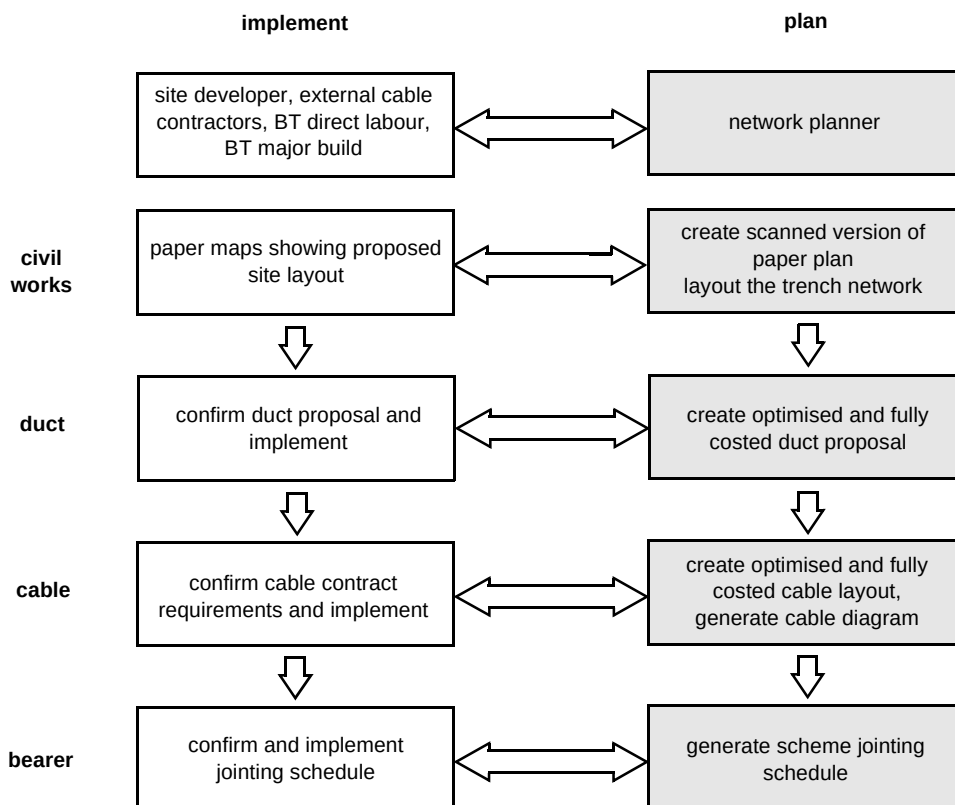


Fig 14 Automating the planning process.

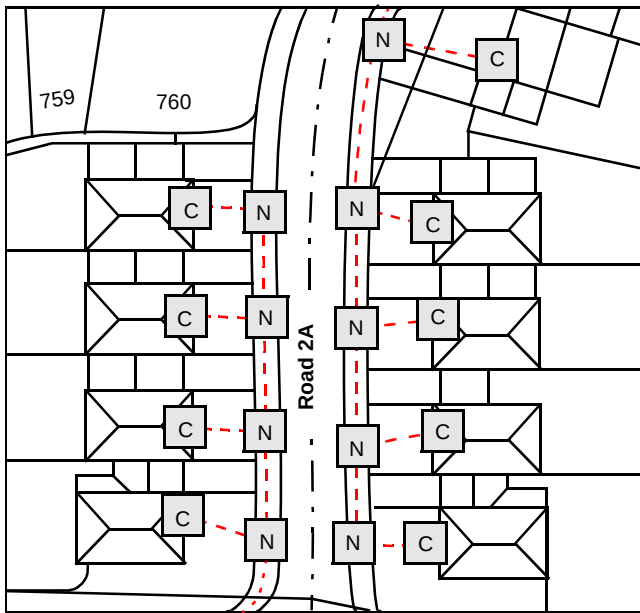


Fig 15 Screen shot of civil works layout illustrating frontage nodes.

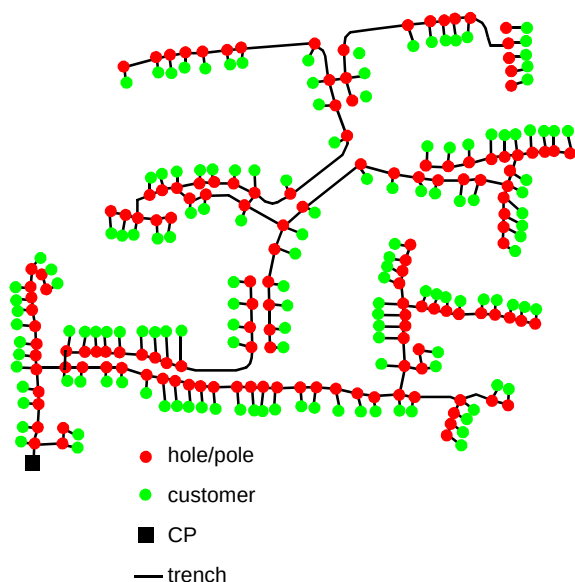


Fig 16 Complete scheme overview highlighting unconnected civil works layer.

Upon completion of the design, a detailed breakdown of costs, bill of materials, cable and bearer documentation can be generated. The cable diagrams created by the system mirror the actual structure of the cable layer of the network and therefore save significant time and errors in translating from cable runs into a cable diagram. Similarly jointing information can be created based on the relative positions of DPs and customers to the PCP.

Figures 17 and 18 respectively illustrate cable and jointing schedule documentation generated by the system.

7. Business benefits and productivity gains

The productivity improvements gained through using the modular, PC-based planning tools described above can be dramatic. For example, the manual production of a financial approval case (FAC) for a sizeable business park would take approximately 2 man-days. When automated via TASC, the job is completed in around 15 minutes.

Results for the GenOSys network optimiser tool have been equally impressive. GenOSys has been constructed, tested and deployed in national roll-out since July 1998 to about 20 planning offices. It has been well received by the BT UK planning community. In the productivity tests at Crayford (North London) the application of the system brought down planning design times for a typical 120 house greenfield scheme from an average of about 20 hours to less than 2 hours. For a 200-house residential scheme, the time to produce a detailed development plan has been reduced from 23 hours to about 3 hours. These results have been highlighted in Fig 19. Interestingly, the main improvement (approximately a factor of 10) comes about even when using the tool in 'manual' mode (i.e. no automatic optimisation) because the GIS allows the planner to quickly lay out the network, cost it, and produce the necessary maps and documents (e.g. bill of materials) for the network builders. Automatic network optimisation offers an additional improvement in guaranteeing the cost optimisation (typically 20% better than an average planner), but the main benefits of automatic optimisation are greater standardisation, consistency and quality, even with less-skilled human operators.

These performance improvements in the planning process are being implemented by relatively small systems, developed quickly, in close association with end-users. The rapid development cycle allows various experimental concepts or prototype tools to be quickly tested in use, and functionality to be modified or discarded in response. This obviates the need for an extensive customer requirements capture phase, and can make redundant the need for the detailed analysis of existing manual processes, that is sometimes required in conventional business process re-engineering.

8. Conclusions

In conclusion, it can be seen that new methods and approaches to software development can dramatically reform important aspects of network planning and management. PC power and the rapid development of PC-based, modular software tools, sharing core data with larger systems, is a quick and effective way to improve planning productivity. Productivity increases by a factor of between 10 and 200 have been achieved from 'smart' PC-based tools developed relatively quickly and cheaply within BT. Such tools also allow the simultaneous consideration of all the

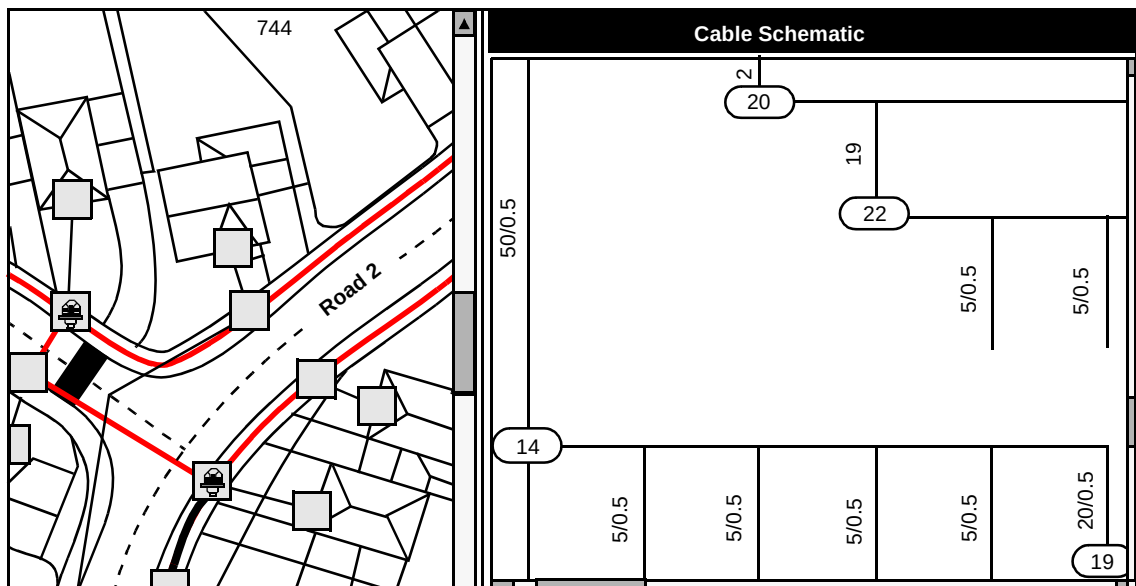


Fig 17 Cable schematic of optimised plan.

Scheme Joining Schedule (A154)							
Job description:							
Planned by:							
Exchange area:							
Estimate number:							
MDF			PCP			DP	
MDF no	bar number	pair number	PCP number	E-side term	D-side term	DP number	pair number
MDF1	AA	20	PCP48	200	400	2	1
MDF1	AA	21	PCP48	201	401	2	2
MDF1	AA	22	PCP48	202	402	2	3
MDF1	AA	23	PCP48	203	403	2	4

Fig 18 Scheme joining schedule.

elements of 'total' network performance, and optimal balances of function, reliability and cost can be achieved using advanced optimisation techniques. The relatively low cost and high speed of development of these tools calls into question the need for extensive re-engineering analyses of existing processes, where the danger is 'paralysis by analysis', and challenges the cost-efficiency of 'big systems'. It is often more effective to rapidly synthesise a new automated process, by developing a small application and testing its effect, before moving on to treat the next obvious productivity problem.

If the key requirements of modularity and data-sharing are adhered to, the result will be a cohesive set of network planning tools, offering massive planning productivity improvements, the better integration of network planning and management, and the guarantee of optimally designed, high-performance networks.

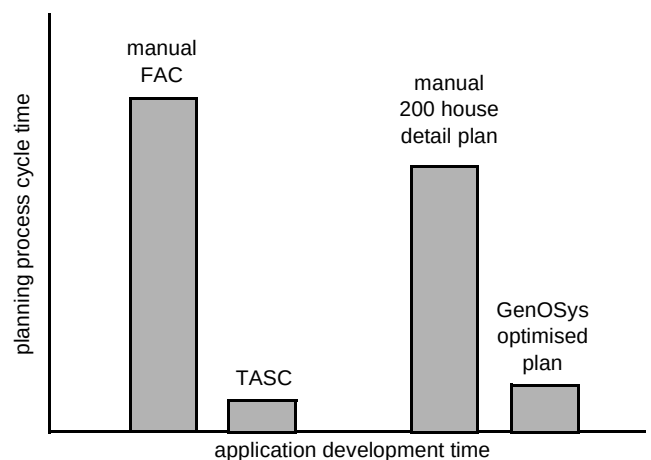


Fig 19 Productivity impact of RAD automatic planning tools.

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